Math Camp

Lecture 2

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Vector spaces

- A *linear combination* of vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n$ has the form

$$c_1 \mathbf{v}_1 + \ldots + c_n \mathbf{v}_n$$

  where $c_1, \ldots, c_n \in \mathbb{R}$ are the combination's coefficients
- A *vector space* means a collection in which any linear combination is sensible
- A vector space must have at least one element, its zero vector $\mathbf{0}$
  - Thus a one-element vector space is the smallest possible

Vector spaces

- A *vector space* (over $\mathbb{R}$) consists of a set $\mathbf{V}$ along with two operations $+$ (vector addition) and $\cdot$ (scalar multiplication) subject to the conditions for all vectors $\mathbf{v}, \mathbf{w}, \mathbf{u} \in \mathbf{V}$ and all scalars $r, s \in \mathbb{R}$

  1. $\mathbf{V}$ is closed under operations,
     - $\mathbf{v} + \mathbf{w} \in \mathbf{V}$
     - $r \cdot \mathbf{v} \in \mathbf{V}$
  2. $+$ is commutative, $\mathbf{v} + \mathbf{w} = \mathbf{w} + \mathbf{v}$
  3. $+$ and $\cdot$ are associative
     - $(\mathbf{v} + \mathbf{w}) + \mathbf{u} = \mathbf{v} + (\mathbf{w} + \mathbf{u})$
     - $r \cdot (s \cdot \mathbf{v}) = (r \cdot s) \cdot \mathbf{v}$

Vector spaces

4. $+$ and $\cdot$ are distributive
   - $r \cdot (\mathbf{v} + \mathbf{w}) = r \cdot \mathbf{v} + r \cdot \mathbf{w}$
   - $(r + s) \cdot \mathbf{v} = r \cdot \mathbf{v} + s \cdot \mathbf{v}$
5. Additive identity, $\exists \mathbf{0} \in \mathbf{V}$ such that $\mathbf{v} + \mathbf{0} = \mathbf{0} + \mathbf{v} = \mathbf{v}$
6. Multiplicative identity, $1 \cdot \mathbf{v} = \mathbf{v}$ for $1 \in \mathbb{R}$
7. Additive inverse, $\forall \mathbf{v} \exists -\mathbf{v}$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$

Vector spaces

Is $\mathbb{R}$ a vector space?

Vector spaces

- If $\mathbf{V}$ and $\mathbf{W}$ are vector spaces, $\mathbf{V} \oplus \mathbf{W}$ is the Cartesian product of $\mathbf{V}$ and $\mathbf{W}$, called the *direct sum*.
- $\mathbf{x}, \mathbf{y} \in \mathbf{V} \oplus \mathbf{W}$ implies $\mathbf{x} = (\mathbf{v}_1, \mathbf{w}_1)$ and $\mathbf{y} = (\mathbf{v}_2, \mathbf{w}_2)$, for some $\mathbf{v}_1, \mathbf{v}_2 \in \mathbf{V}$ and $\mathbf{w}_1, \mathbf{w}_2 \in \mathbf{W}$
- Addition is coordinate-wise, $\mathbf{x} + \mathbf{y} = (\mathbf{v}_1 + \mathbf{v}_2, \mathbf{w}_1 + \mathbf{w}_2)$
- So is scalar multiplication, $r \cdot \mathbf{x} = (r \cdot \mathbf{v}_1, r \cdot \mathbf{w}_1)$
- $\mathbf{V} \oplus \mathbf{W}$ is itself a vector space

Vector spaces

Is $\mathbb{R}^2$ a vector space?

Subspaces and Spans

- For any vector space $\mathbf{V}$, a *vector subspace* $\mathbf{W} \subseteq \mathbf{V}$ is itself a vector space under the inherited operations
  - Equivalently, $\mathbf{W} \neq \varnothing$ and $\mathbf{W}$ is closed under linear combinations

- Example: $\mathbf{W}_1$, a set of all vectors in $\mathbb{R}^2$ whose components sum to zero

Subspaces and Spans

- The *span* or *linear closure* of a nonempty subset $A \subseteq \mathbf{V}$ is the set of all finite linear combinations of vectors from A

$$\text{span}(A) = \{c_1\mathbf{a}_1 + ... + c_n\mathbf{a}_n \mid (c_1, ..., c_n \in \mathbb{R} \text{ and } \mathbf{a}_1, ..., \mathbf{a}_n \in A)\}$$

  - The span of any subset is a subspace
  - span(A) is the smallest subspace of $\mathbf{V}$ containing all of the members of A
  - $\{(1,0,0)\} \subseteq \mathbb{R}^3$, span$\{(1,0,0)\}$ is the x-axis
  - span$\{(1,0,0),(0,1,0)\}$ is the xy-plane
  - span$\{(1,0,0),(2,0,0)\}$ is the x-axis

Linear Independence

- In any vector space $\mathbf{V}$, a set of vectors is *linearly independent* if none of its elements is a linear combination of the others from the set

$$\{\mathbf{v}_1, ..., \mathbf{v}_n\} \subseteq \mathbf{V} \text{ and } \sum_{k=1}^{n} c_k \mathbf{v}_k = \mathbf{0} \Rightarrow c_k = 0 \forall k = 1, ..., n$$

  - Otherwise the set is linearly dependent

- $\{(1, 0, 0), (0, 1, 0)\}$ is linearly independent
- $\{(1, 0, 0), (2, 0, 0)\}$ is linearly dependent

Linear Independence

- A *basis* of $\mathbf{V}$ is a linearly independent set $B \subseteq \mathbf{V}$ that spans the space $\mathbf{V}$
- In any $\mathbf{V}$, $B \subseteq \mathbf{V}$ is a basis iff each $\mathbf{v} \in \mathbf{V}$ can be expressed as a linear combination of the elements of B in only one way
- For any $\mathbb{R}^n$,

$$\mathcal{E} = \left\langle \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix} \right\rangle$$

  is the standard basis. We denote these vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$
- In calculus, $\mathbf{i}$ and $\mathbf{j}$ denote the standard basis vectors of $\mathbb{R}^2$ instead of $\mathbf{e}_1$ and $\mathbf{e}_2$

Linear Independence

Standard and non-standard basis for $\begin{pmatrix} 3 \\ 5 \\ -2 \end{pmatrix}$

Linear Independence

Steinitz Exchange Lemma

- If $A = \{\mathbf{a}_1, ..., \mathbf{a}_n\} \subseteq \mathbf{V}$ is a basis for $\mathbf{V}$ and $B = \{\mathbf{b}_1, ..., \mathbf{b}_m\} \subseteq \mathbf{V}$ is linearly independent then $m \leq n$
- **Corollary**: In any finite-dimensional vector space, all bases have the same number of elements

Linear Independence

Exchange Theorem

- Assume that $B = \langle \beta_1, ..., \beta_n \rangle$ is a basis for $\mathbf{V}$ and that $\mathbf{v} \in \mathbf{V}$ has the representation

$$\mathbf{v} = \sum_{i=1}^{n} c_i \beta_i$$

  - For any j with $c_j \neq 0$, exchanging β_j for $\mathbf{v}$ yields another basis for the space

Dimension

- The *dimension* of a vector space, is the number of vectors in any of its bases
  - If $B \subseteq \mathbf{V}$ is a basis for $\mathbf{V}$ then $\dim \mathbf{V} = |B|$
- Let $\dim \mathbf{V} = n$. Then,
  - If $B \subseteq \mathbf{V}$ is linearly independent then B is a basis for $\operatorname{span}(B)$ and $\dim \operatorname{span}(B) = |B|$
  - If $B \subseteq \mathbf{V}$ is linearly independent and $|B| = m < n$ then $\exists C \subseteq \mathbf{V}$ so that $B \cup C$ is a basis for $\mathbf{V}$
  - If $\mathbf{W}$ is a subspace of $\mathbf{V}$ then $\dim \mathbf{W} \leq \dim \mathbf{V}$

Linear Functions

- **Category Theory** is a general theory of mathematical structures and their relations
  - A *category* is a collection of *objects* and *arrows* (*morphisms*)
    - Objects are the items of interest and arrows are relations between these items

- In the category of linear spaces, the objects are vector spaces and the morphisms are linear maps
  - Linear maps are functions between vector spaces that preserve their vector-space (i.e. linear) structure

Linear Functions

- If $\mathbf{V}$ and $\mathbf{W}$ are vector spaces then a function $f : \mathbf{V} \to \mathbf{W}$ that preserves addition

$$\text{if } \mathbf{v}_1, \mathbf{v}_2 \in \mathbf{V} \text{ then } f(\mathbf{v}_1 + \mathbf{v}_2) = f(\mathbf{v}_1) + f(\mathbf{v}_2)$$

  and scalar multiplication

$$\text{if } \mathbf{v} \in \mathbf{V} \text{ and } r \in \mathbb{R} \text{ then } f(r \cdot \mathbf{v}) = r \cdot f(\mathbf{v})$$

  is a *homomorphism* or *linear map*

  - A linear map from a vector space $\mathbf{V}$ into itself $t : \mathbf{V} \to \mathbf{V}$ is a *linear transformation*

Linear Functions

- The map f is an *isomorphism* if it is invertible, i.e, there exists a linear map $g : \mathbf{W} \to \mathbf{V}$ such that

$$f(g(\mathbf{w})) = \mathbf{w} \quad \text{and} \quad g(f(\mathbf{v})) = \mathbf{v}$$

  for all $\mathbf{v} \in \mathbf{V}$ and $\mathbf{w} \in \mathbf{W}$
  - Invertibility means that f is one-to-one and onto
  - If such an isomorphism exists, we say that $\mathbf{V}$ and $\mathbf{W}$ are *isomorphic*, $\mathbf{V} \simeq \mathbf{W}$

Kernel and Range

- The *range* of a linear map $f : \mathbf{V} \to \mathbf{W}$ is

$$f(\mathbf{V}) = \{f(\mathbf{v}) \mid \mathbf{v} \in \mathbf{V}\} \subseteq \mathbf{W}$$

  - The range of f is the image of $\mathbf{V}$ in $\mathbf{W}$ under the action of f
  - The range of f is a subspace of $\mathbf{W}$
  - The dimension of a range is the map's *rank*

- The *kernel* or *null space* of a linear map $f : \mathbf{V} \to \mathbf{W}$ is the inverse image of $\mathbf{0_W}$

$$\ker(f) = f^{-1}(\mathbf{0_W}) = \{\mathbf{v} \in \mathbf{V} \mid f(\mathbf{v}) = \mathbf{0_W}\}$$

  - The kernel of f is the collection of vectors that f sends to zero
  - The kernel of f is a subspace of $\mathbf{V}$
  - The dimension of the kernel is the map's *nullity*

Kernel and Range

Rank Nullity Theorem

If $\mathbf{V}$ is finite dimensional and $f : \mathbf{V} \to \mathbf{W}$ is a linear map then the rank of f plus the nullity of f equals the dimension of its domain $\mathbf{V}$

$$\dim \ker(f) + \dim f(\mathbf{V}) = \dim \mathbf{V}$$

- This theorem summarizes the structural information that is lost by a map f
  - If $\mathbf{v}_1, \mathbf{v}_2 \in \ker(f)$ then $f(\mathbf{v}_1) = f(\mathbf{v}_2)$, i.e, f does not distinguish $\mathbf{v}_1$ and $\mathbf{v}_2$
  - If $\mathbf{v}_1 \in \ker(f)$ and $\mathbf{v}_2 \notin \ker(f)$ then $f(\mathbf{v}_2) = f(\mathbf{v}_1 + \mathbf{v}_2)$, i.e, f does not distinguish $\mathbf{v}_2$ and $\mathbf{v}_1 + \mathbf{v}_2$

Kernel and Range

- For $D : \mathscr{P}_3 \to \mathscr{P}_2$
  - $\ker(D)$ is the constants, with nullity 1
  - The range is all of $\mathscr{P}_2$ with rank 3
  - $\dim \mathscr{P}_3 = 1 + 3 = 4$
- For $f : \mathbb{R}^4 \to \mathbb{R}^3$ such that $f(x_1, x_2, x_3, x_4) = (x_1 - x_3, x_2 - x_4, 0)$
  - $\ker f = \text{span}\{(1, 0, 1, 0), (0, 1, 0, 1)\}$, with nullity 2
  - $f(\mathbb{R}^4) = \{(\alpha, \beta, 0)\} = \text{span}\{(1, 0, 0), (0, 1, 0)\}$, with rank 2
  - $\dim \mathbb{R}^4 = 2 + 2 = 4$

Kernel and Range

- The behavior of a linear map is entirely characterized by its behavior on a basis
  - If $A = \{\mathbf{a}_1, ..., \mathbf{a}_n\}$ is a basis for $\mathbf{V}$ and $\{\mathbf{w}_1, ..., \mathbf{w}_n\}$ are not-necessarily linearly independent, or even distinct vectors in $\mathbf{W}$ then $f : \mathbf{V} \to \mathbf{W}$ can be defined as sending $\mathbf{a}_i$ to $\mathbf{w}_i$, and extending linearly,

$$\mathbf{v} = \sum_{k=1}^{n} c_k \mathbf{a}_k \to \sum_{k=1}^{n} c_k \mathbf{w}_k = f(\mathbf{v})$$

- If $\dim(\mathbf{V}) = \dim(\mathbf{W}) = n$, then $\mathbf{V}$ is isomorphic to $\mathbf{W}$